

DATA FLOW DIAGRAMS (DFDs)

Company Internal Reporting System

Prepared By	Mugisha Yves
Document	DFDs

1. Introduction to DFDs

Data Flow Diagrams (DFDs) illustrate how data flows through the Company Internal Reporting System. They show what data enters and leaves the system, what processes transform that data, and where data is stored. DFDs focus purely on data movement — not on implementation details.

1.1 DFD Notation Key

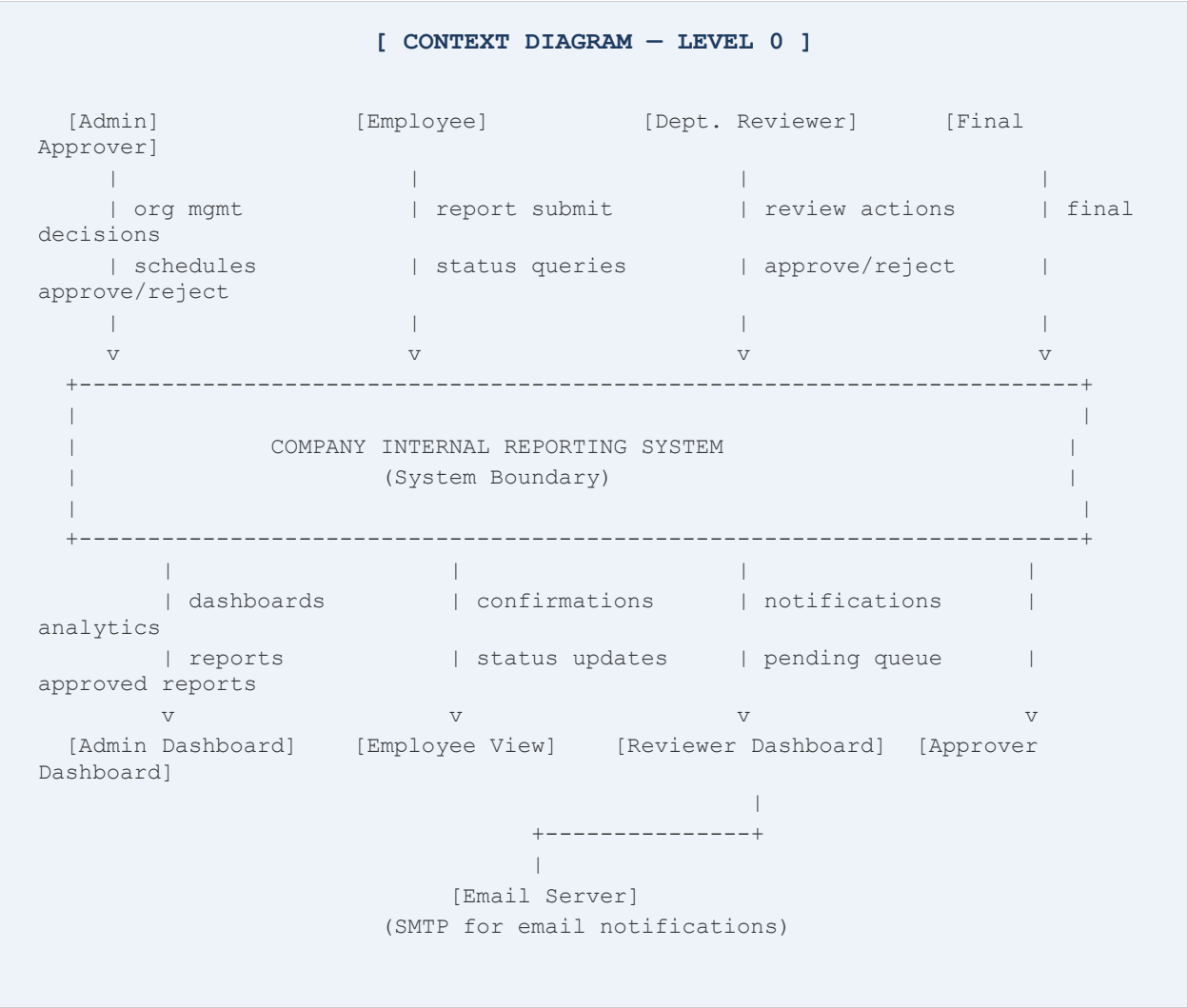
[Ext]	External Entity — An actor outside the system boundary (e.g., Employee, Admin, Email Server).
(Proc)	Process — A system function that transforms or acts on data (e.g., Authenticate User, Submit Report).
= DB =	Data Store — Where data is persisted in the database (e.g., Users, Reports, Notifications).
-->	Data Flow — Arrow showing direction of data movement, labeled with what data flows.

1.2 DFD Levels

- Context Diagram (Level 0) – Highest-level: entire system as one process with all external entities.
- Level 1 DFD – Expands the system into major functional processes and data stores.
- Level 2 DFDs – Breaks down key Level 1 processes (P4, P5, P6, P8) into detailed sub-processes.

2. Context Diagram (Level 0)

The Context Diagram represents the entire system as a single black-box process. It defines the system boundary and shows all external entities that interact with the system.



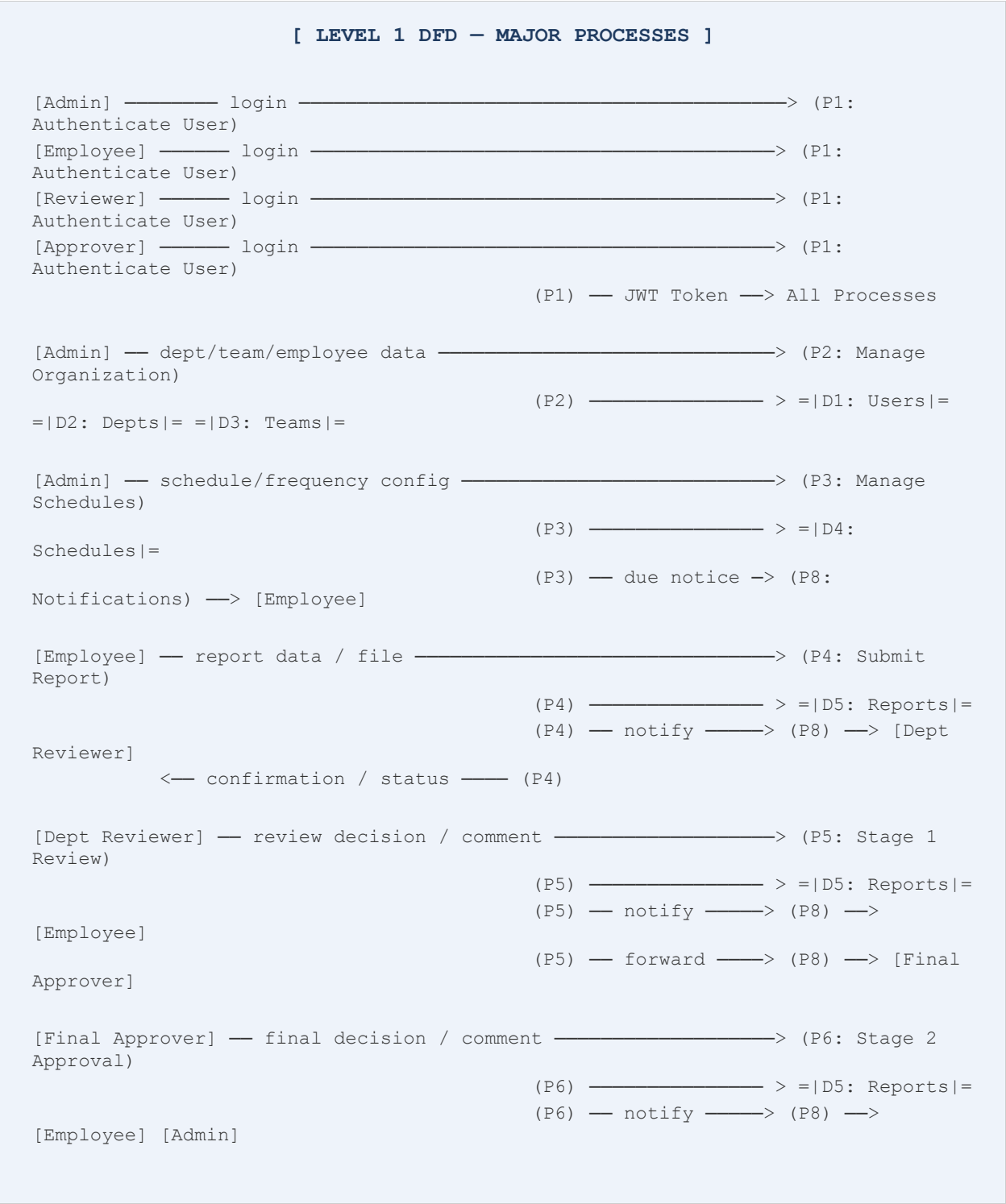
2.1 External Entities and Data Flows

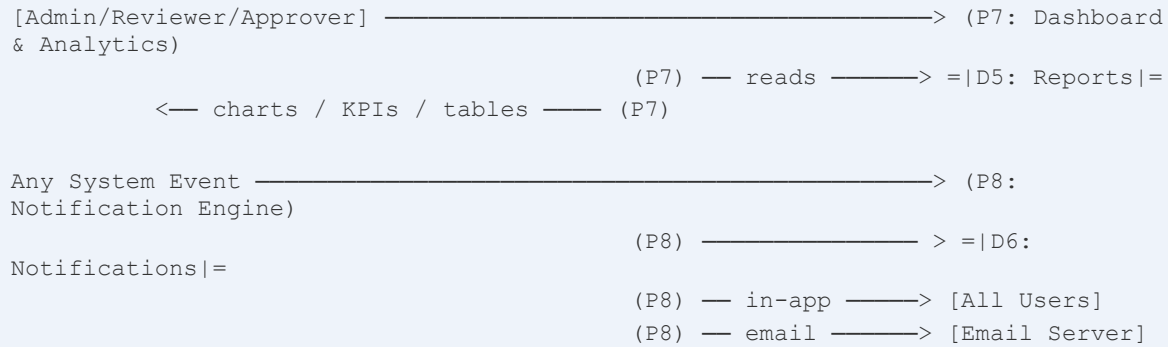
Entity	Data Sent INTO System	Data Received FROM System
Admin	Login credentials, dept/team data, employee registrations, report schedules	Dashboards, org confirmations, compliance reports
Employee	Login credentials, report form data, file uploads	Submission confirmations, status updates, notifications

Entity	Data Sent INTO System	Data Received FROM System
Department Reviewer	Login, review decisions (Approve/Reject/Changes), comments	Pending reports queue, submission notifications, confirmations
Final Approver (COO)	Login, final decisions, mandatory comments	Stage-1 approved reports, analytics, notifications
Email Server (SMTP)	N/A — External outbound service	Formatted email messages triggered by system events

3. Level 1 DFD

The Level 1 DFD decomposes the single system process into 8 major functional processes, showing the data stores and data flows between processes and external entities.





3.1 Process Descriptions

ID	Process	Input	Output	Data Store(s)
P1	Authenticate User	Username, password	JWT token or error	D1: Users
P2	Manage Organization	Dept/team/employee details	Org records, confirmations	D1, D2, D3
P3	Manage Schedules	Frequency, type, dept/team, deadline	Timelines, reminders	D4: Schedules
P4	Submit Report	Report form data, file attachment	Report record, confirmation	D5: Reports
P5	Stage 1 Review	Review action, comment	Updated status, notification event	D5: Reports
P6	Stage 2 Approval	Final decision, comment	Final status, notification event	D5: Reports
P7	Dashboard & Analytics	Query filters	KPI charts, compliance metrics	D5: Reports
P8	Notification Engine	System event trigger	In-app alerts, emails	D6: Notifications

3.2 Data Stores

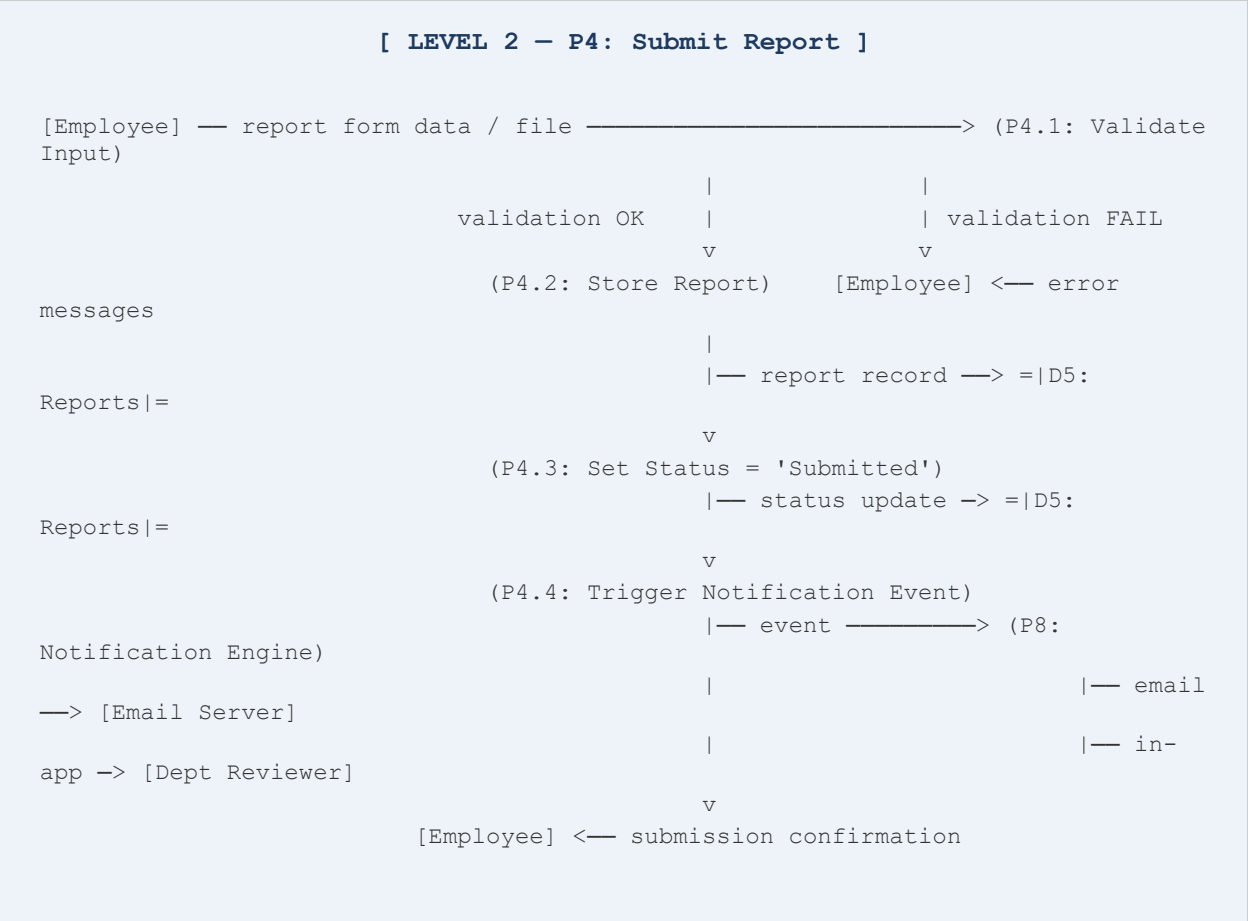
ID	Name	Contents
D1	Users	user_id, name, email, password_hash, role, dept_id, team_id, created_at
D2	Departments	dept_id, name, reviewer_id, description, created_at
D3	Teams	team_id, name, dept_id, created_at

ID	Name	Contents
D4	Schedules	schedule_id, report_type, frequency, start_date, deadline, dept_id, team_id
D5	Reports	report_id, employee_id, schedule_id, content, file_path, status, review_history, timestamps
D6	Notifications	notif_id, user_id, event_type, message, is_read, created_at

4. Level 2 DFDs

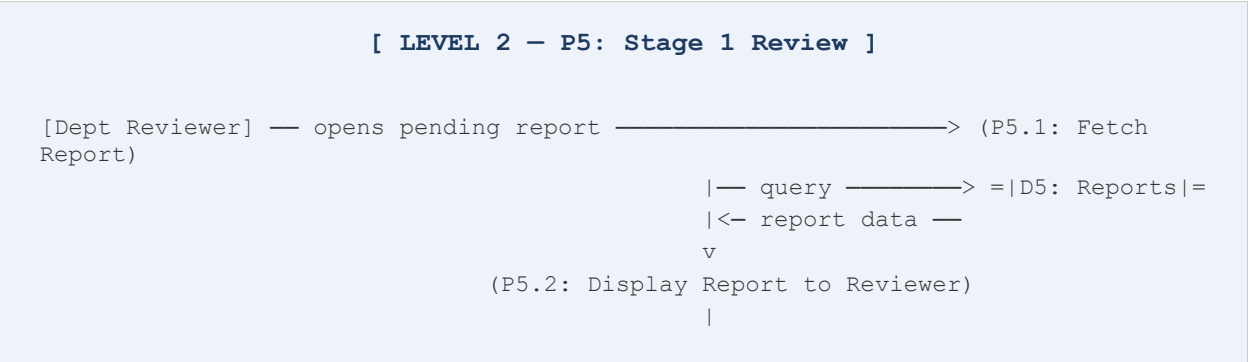
4.1 Level 2 — P4: Report Submission

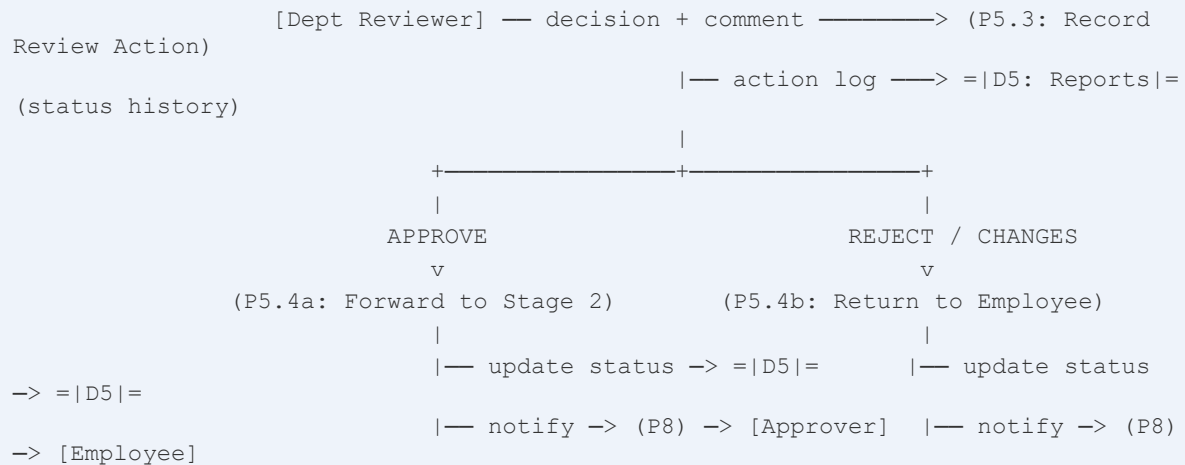
Breaks down the Report Submission process into its detailed sub-steps.



4.2 Level 2 — P5: Stage 1 Review

Breaks down the Department Review process into its detailed sub-steps.

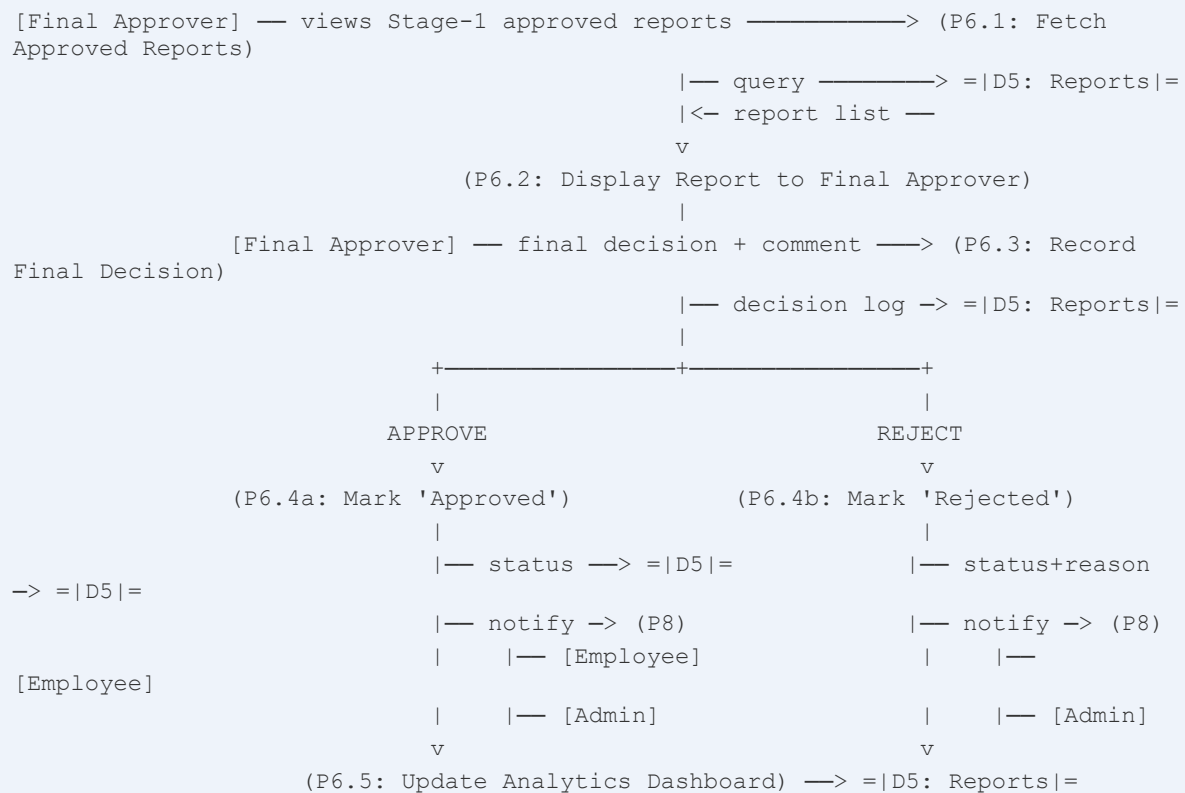




4.3 Level 2 — P6: Stage 2 Final Approval

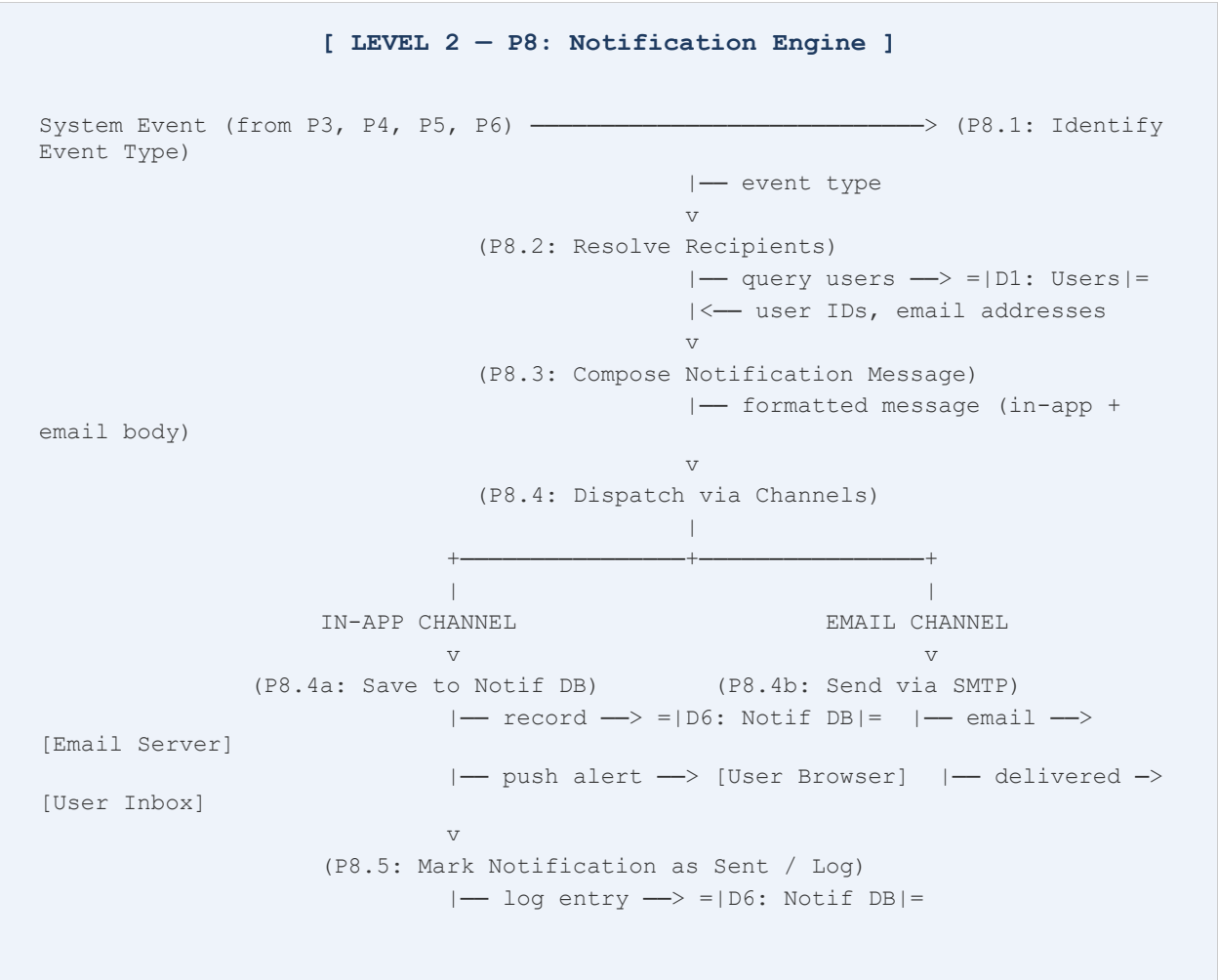
Breaks down the Final Approval process into its detailed sub-steps.

[LEVEL 2 — P6: Stage 2 Final Approval]



4.4 Level 2 — P8: Notification Engine

Breaks down the Notification Engine process triggered by any system event.



5. Data Flow Summary

All significant data flows across the system are summarized below:

Flow ID	From	To	Data Description
DF-01	Admin	P2: Manage Org	Employee name, email, role, dept, team assignment
DF-02	P2	D1/D2/D3	Persisted user, department, team records
DF-03	Admin	P3: Schedules	Report type, frequency, start date, deadline, dept/team
DF-04	P3	D4: Schedules	Schedule record with auto-generated timeline
DF-05	P3	P8: Notifications	Report-due event trigger with employee targets
DF-06	Employee	P4: Submit	Report form fields and/or uploaded file (PDF/DOCX/XLSX)
DF-07	P4	D5: Reports	New report record with status = 'Submitted'
DF-08	P4	P8: Notifications	Submission event for Department Reviewer
DF-09	Dept Reviewer	P5: Review	Decision (Approve/Reject/Changes), optional comment
DF-10	P5	D5: Reports	Updated status, review log entry with timestamp
DF-11	P5	P8: Notifications	Review outcome event for Employee / Final Approver
DF-12	Final Approver	P6: Approval	Final decision (Approve/Reject), mandatory comment
DF-13	P6	D5: Reports	Final status, audit log entry
DF-14	P6	P8: Notifications	Final outcome event for Employee and Admin
DF-15	P8	D6: Notifications	Notification record (user, message, read status)
DF-16	P8	Email Server	Formatted email message via SMTP
DF-17	D5: Reports	P7: Dashboard	Aggregated report data, counts, status breakdowns
DF-18	P7: Dashboard	Admin/Reviewer/Approver	KPI charts, compliance metrics, filterable report tables